

Sales Performance Analysis

A project designed to analyze customer profiles, loyalty levels, and demographic trends using structured sales and customer data.

Description

This project provides insights into customer behavior by combining demographic attributes (age, gender, location) with loyalty segmentation (Platinum, Gold, Silver, Bronze, Unknown). It helps businesses understand their customer base, identify high-value segments, and design targeted marketing strategies. The dataset includes over 1,400+ customer records with detailed attributes such as **Customer_ID, Name, Age, Gender, City, State, Country, and Loyalty_Level**.

Getting Started

1. Descriptive Analysis (What Happened?)

Descriptive analysis summarizes the overall business performance from the dataset.

- The dataset contains **2,000 sales transactions** across multiple stores, products, and customers.
- Total revenue generated exceeded **₹25 lakh**, showing strong sales activity.
- The average order value indicates customers are making moderate-value purchases.
- Multiple payment methods were used, with **PayPal and Credit Card** contributing the highest revenue.
- Store-wise analysis shows that a few stores contribute a larger share of total sales compared to others.
- Product performance varies significantly, with some products generating consistently higher revenue than others.
- Discounts were applied on most transactions, helping drive customer purchases.

2. Diagnostic Analysis (Why Did It Happen?)

Diagnostic analysis identifies the reasons behind sales performance.

- Stores with higher revenue also recorded higher transaction volumes, indicating strong customer traffic.
- Products with competitive pricing and customer demand contributed the most revenue.

- Discount offers positively influenced sales volume, especially for frequently purchased products.
- Loyal customers generated repeat purchases, increasing overall revenue contribution.
- Payment method preferences affected sales distribution, with digital payments being more popular than cash-based transactions.
- Regional and store-level variations suggest that customer buying behavior differs across locations

3. Predictive Analysis (What Is Likely to Happen?)

Predictive analysis uses historical trends to forecast future outcomes.

- High-performing products are likely to continue generating strong revenue if demand remains stable.
- Stores currently leading in sales are expected to maintain growth due to consistent customer activity.
- Digital payment adoption is likely to increase further, reducing dependence on cash transactions.
- Loyalty program customers are expected to contribute more repeat purchases in future periods.
- Seasonal demand patterns may create revenue spikes for specific product categories.

4. Prescriptive Analysis (What Should Be Done?)

Prescriptive analysis provides recommendations for improving business performance.

- Increase inventory levels for top-selling products to avoid stock shortages.
- Expand promotional campaigns in high-performing stores to maximize revenue.
- Offer personalized discounts and rewards to loyalty customers to encourage repeat purchases.
- Introduce targeted marketing for low-performing products and regions.
- Promote digital payment incentives to improve customer convenience and transaction speed.
- Review pricing and discount strategies regularly to balance profitability and sales growth.
- Replicate successful sales practices from top-performing stores across other locations.

Dashboard Insights

KPI Cards

- Total Revenue
- Total Orders
- Average Order Value
- Average Discount %

Pivot Table Insights

1. **Store vs Revenue** – Identifies top-performing stores.
2. **Category vs Revenue** – Shows highest earning product categories.
3. **Payment Type vs Amount** – Highlights customer payment preferences.
4. **Gender vs Customer Count** – Reveals customer demographics.
5. **Store vs Discount** – Measures impact of discounts on sales.

Dashboard Conclusion

The dashboard reveals that revenue is concentrated among a few high-performing stores and products. Digital payment methods dominate transactions, while loyalty customers contribute significantly to sales. By optimizing inventory, expanding successful promotions, and strengthening customer retention strategies, the business can improve future revenue and operational efficiency.